

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

3.41. **Pushing on a mass** **

Name:

Section:

- (a) A mass m starts at rest. You push on it with a constant force F . How much time t does it take for the mass to move a distance x ? (Both t and x here are measured in the lab frame.)
- (b) After a very long time, m 's speed will approach c . It turns out that it approaches c sufficiently fast so that after a very long time, m will remain (approximately) a constant distance (as measured in the lab frame) behind a photon that was emitted at $t = 0$ from the starting position of m . What is this distance? (The Taylor series $\sqrt{1 + \epsilon} \approx 1 + \epsilon/2$ will be helpful.)

1. **Relativistic force.** (50 points) Morin exercise 3.41 (p. 158). For part (a), explain your result in the ultrarelativistic limit $x \rightarrow \infty$ and the non-relativistic limit $x \ll mc^2/F$. *Hint: for part (a), use $\Delta E = F\Delta x$ and $F = dp/dt$, as well as the relation $E^2 - p^2c^2 = m^2c^4$ from Discussion 8, to equate two expressions for the momentum in order to solve for t .*

- (a) The kinetic energy of the mass after it has traveled a distance x is $\Delta E = Fx$, so its total energy is $E = mc^2 + Fx$. Its momentum is then

$$p = \frac{1}{c}\sqrt{E^2 - m^2c^4} = \frac{1}{c}\sqrt{(mc^2 + Fx)^2 - m^2c^4} = \frac{1}{c}\sqrt{F^2x^2 + 2mc^2Fx} = \sqrt{F^2x^2/c^2 + 2mFx}.$$

On the other hand, since the force is constant, we can integrate $F = dp/dt$ to get $p(t) = Ft$. Setting these two expressions for p equal gives

$$t = \frac{\sqrt{F^2x^2/c^2 + 2mFx}}{F}.$$

In the limit $x \rightarrow \infty$, we have $t \approx x/c$, so the object is traveling approximately at the speed of light. In the limit $x \ll mc^2/F$, we can Taylor-expand the square root, first factoring out $\sqrt{2mx/F}$, to find

$$t = \sqrt{2mx/F} \sqrt{1 + \frac{Fx}{2mc^2}} = \sqrt{2mx/F} \left(1 + \mathcal{O}\left(\frac{x}{mc^2/F}\right) \right)$$

The first term in parentheses is the non-relativistic result from combining $x = at^2/2$ and $F = ma$, and the next term is the first relativistic correction.

- (b) This time we factor x/c out of the square root to get

$$t = \frac{x}{c} \sqrt{1 + \frac{2mc^2}{Fx}}$$

Taylor-expanding this gives

$$t \approx \frac{x}{c} \left(1 + \frac{mc^2}{Fx} \right) = \frac{x}{c} + \frac{mc}{F}.$$

The position of the photon at time t is $x' = ct = x + \frac{mc^2}{F}$, while the mass is at position x . So the mass is behind the photon by a distance mc^2/F .

2. **Relativistic rockets and practice with separation of variables.** Suppose a rocket of mass M ejects propellant at a constant velocity w with respect to the rocket, thus achieving an instantaneous velocity U with respect to the center of mass of the rocket-propellant system.

- (a) (30 points) Consider an infinitesimal amount of propellant with mass dm expelled from the rocket. Use conservation of energy, conservation of momentum, and velocity addition to derive the relativistic rocket equation:

$$\frac{dM}{M} = -\frac{dU}{w(1 - U^2/c^2)}. \quad (1)$$

Feel free to refer to Morin Sec. 3.6 for inspiration, but note the difference here: our propellant is massive and has velocity $w < c$. When $w = c$, Eq. (1) reduces to Morin (3.72). (The Newtonian (non-relativistic) rocket equation is obtained by Taylor-expanding the denominator in U , giving $dM/M = -dU/w$.) *Hint: your algebra will be MUCH simpler if you start your calculation in the rest frame of the rocket, where the rocket starts with speed 0 and ends up with speed dv after ejecting the propellant. Remember that mass is not conserved ($dm \neq -dM$), but energy is. Relate dv to dM and then use velocity addition to relate dU to dv .*

The calculation is easiest in the rocket's rest frame, where the rocket starts at rest and ends up with speed dv . Note that the mass lost by the rocket $-dM$ is *not* equal to the mass expelled dm , since mass is not conserved in relativity. However, energy is conserved, so we can use conservation of energy to relate dm to dM , conservation of momentum to relate dM to dv , and finally velocity addition to relate dv to dU . We will also set $c = 1$ for simplicity until the end of the calculation. Conservation of energy gives

$$M = \gamma_w dm + \gamma_v (M + dM).$$

But $\gamma_v = 1 + \mathcal{O}(dv^2)$, so to leading order we can set $\gamma_v = 1$. Thus $dM = -\gamma_w dm$. Conservation of momentum gives

$$0 = -\gamma_w w dm + \gamma_v (M + dM) dv.$$

We can set $\gamma_v = 1$ and drop $dM dv$ which is second order in differentials. Using our conservation of energy result, we have simply $dv = -w dM/M$. To find the rocket's velocity, we assume it has velocity U with respect to the center of mass (or some other frame, like the ground) and say that its new velocity is $U + dU$. To get from the rocket's rest frame to the ground frame, we use velocity addition, in the form used in lecture where we Taylor-expanded the denominator:

$$U + dU = \frac{U + dv}{1 + U dv} \approx (U + dv)(1 - U dv) \approx U + (1 - U^2) dv.$$

Thus $dU = (1 - U^2) dv$, and plugging in $dv = -w dM/M$, we obtain the desired rocket equation once we replace U^2 with U^2/c^2 in the denominator.

- (b) (20 points) Suppose the initial mass of the rocket is M_0 and the initial velocity is $U_0 = 0$. Integrate Eq. (1) using separation of variables to show that the mass of the rocket as a function of its velocity is

$$M(U) = M_0 \left[\frac{1 - U/c}{1 + U/c} \right]^{\frac{c}{2w}} \quad (2)$$

This again reduces to Morin's (3.73) when $w = c$. You can use Morin's derivation on p. 147 to do the integral.

Using Morin (3.81) and the fact that w is constant, we integrate the right-hand side of the rocket equation to get

$$-\frac{1}{w} \int_0^U \frac{dU'}{1 - U'^2/c^2} = -\frac{c}{w} \ln \left(\sqrt{\frac{1 + U/c}{1 - U/c}} \right) = \ln \left[\left(\frac{1 - U/c}{1 + U/c} \right)^{c/2w} \right],$$

using the fact that $a \ln b = \ln b^a$. Integrating the left-hand side just gives $\int_{M_0}^M \frac{dM'}{M'} = \ln(M/M_0)$. Exponentiating both sides gets rid of the logs and gives the desired mass equation.